

Nicolas Dirnegger

📍 Los Angeles, CA, USA ✉ nickeyd01@ucla.edu 📞 Nicolas Dirnegger in nicolas-dirnegger 📧 Nickeyd01

Education

University of California, Los Angeles
PhD in Electrical and Computer Engineering
MS in Quantum Science and Technology, Physics

Sep 2022 - Expected Jan 2027

ETH Zurich
BS in Electrical and Computer Engineering

Sept 2019 – Aug 2022

Employment

Nokia Bell Labs
Quantum Networking Intern

Murray Hill, NJ
Jun 2026 – Sep 2026

- Quantum teleportation of entangled photons over optical fibers, studying both polarization and time-bin encodings.
- Setting up Bell-state measurement systems for polarization- and time-bin-entangled photon sources and work on implementing entanglement teleportation over optical fibers using Superconducting Nanowire Single-Photon Detectors (SNSPDs), time taggers, as well as free-running and atomic clocks and time-synchronization equipment.

IBM Research-Quantum
Quantum Hardware Engineer Intern

Yorktown Heights, NY
June 2024 – Sep 2024

- Computational analysis of remote entanglement methods for superconducting qubits with transducer and entanglement distillation. Theoretical analysis, experimental implementation, and high-performance simulations using IBM CCC cluster.
- Implemented Monte Carlo framework to emulate remote entanglement process and compared several entanglement distillation procedure such as EPL, DEJMPS, BBPSW and Chi-Protocol
- Analytically verified that EPL distillation based remote entanglement protocol offers local phase stability

NarangLab
Graduate Student Researcher, Prof. Prineha Narang

Los Angeles, CA
June 2023 – Present

- Proposed superconducting diodes (SDs) as coherent nonreciprocal elements in cQED, demonstrating direction-dependent resonance shifts via asymmetric SQUID flux bias and spectroscopic characterization through input-output theory and Heisenberg-Langevin equations.
- Demonstrated nonreciprocal qubit-qubit coupling mediated by SD, achieving directional half-iSWAP gates with tunable entanglement contrast verified via two-qubit state tomography and CHSH inequality violation.
- Designed and simulated SD geometries in metal for integration into cQED readout architectures, with full-wave EM simulation in HFSS to assess on-chip feasibility and mode structure.
- Proposed a multimode superconducting ring resonator scheme leveraging cross-Kerr nonlinearities in a driven-dissipative framework for sensitive probing of time-reversal symmetry breaking (TRSB) in quantum materials; mapped optimal operating regimes near the bifurcation threshold and quantified SNR robustness under realistic noise conditions via semiclassical and full quantum analysis.
- Theoretical analysis and computational models of entanglement distribution and quantum transduction for microwave and optical systems.

Air Force Research Laboratory
Quantum Co-Op Intern

Rome, NY
June 2023 – Sep 2023

- Device theory, design and simulation of low frequency superconducting ring resonator with Matlab, HFSS, Comsol and LTSpice. Design, composition and measurement of PCB resonator using surface mount and through hole components.
- Qubit characterization with single shot measurement. Measurement of qubit system under external laser drive to establish quasi-particle tunneling and resulting effects on T1, T2 and T2* time

Mesoscopic Optics and Quantum Electronics Laboratory
Graduate Research Assistant, Prof. Chee-Wei Wong

Los Angeles, CA
Sep 2022 – June 2023

- Experimental measurements of on-chip integrated octave spanning frequency comb and soliton generation with dispersive waves. Device design, characterization and measurement of photonic ring resonators retrieving Q values, FSR and Cherenkov radiation. Post-processing of data in Matlab.

Institute of Electromagnetic Fields

Zurich, Switzerland

Undergraduate Research Assistant, Prof. Juerg Leuthold

Sep 2021 – Aug 2022

- 2D and 3D Device Design of efficient photonic and plasmonic grating couplers with FDTD and FME using COMSOL, Lumerical.
- Silicon Nitride Grating Coupler designed with 0.5dB coupling efficiency, experimentally verified to 1.1dB

Skills & Projects

Coding Languages: Python, C++, Java, MATLAB, Mathematica

Languages: German, French, English, Spanish

Lab Experience:

- Design and simulation for microwave and optical devices in Comsol, Lumerical, LTSpice, Quantum Metal and ADS Keysight. Basics of GDS, HFSS & KLayout, Superconducting Qubit Measurement, Readout and Setup.
- Fiber and Free Space Optics, Laser Alignment, Frequency Comb Generation, LabView, RF Engineering, Simulink, PCB Design and Measurement, Soldering.

Software Experience:

- Lumerical, Comsol, LTSpice, Ansys HFSS, Qiskit, QuTip, High Performance Computing, ADS Keysight, Cadence AWR Microwave Office
- Typesetting LaTeX, HTML/CSS, and others.
- Machine Learning Frameworks PyTorch, Deep Learning and Reinforcement Learning

Leadership, Outreach & Involvement

Quantum Device Workshop at UCLA

Los Angeles, CA

Organizer

- Organized workshop on superconducting device design and simulation at UCLA in partnership with USC
- 400+ Attendees, Sponsored through industrial quantum partners such as Google Quantum AI, Rigetti, NVIDIA, Keysight and more
- Coordinated outreach, organization and participation of workshop as well as landing experts from field as speakers from across globe

Quantum Computing Student Association - QCSA

Los Angeles, CA

Board Member

- Nonprofit organization involved in spreading quantum computing across universities in the south California area. Organization of events that include academic and industrial speakers, hackathons, workshops, and research initiatives. [LinkedIn](#) [↗](#).

Aerospace Initiative ETHZ - ARIS

Zurich, Switzerland

Hardware Engineer

- Part of the liquid rocket engine team as a hardware engineer on sensors, circuit design, and electrical wiring.

GVI - Volunteer

Pokhara, Nepal

Volunteer

- Volunteering abroad in Pokhara in Nepal to rebuild and renovate a local school. Social media fundraiser to support the local school system in the Pokhara region.

Publications & Patents

- [1] M. Gabriela Boada, **Dirnegger, N.**, Delgado, A., Narang, P. "An extensive theory of nonlinearly intercoupled pseudomodes for noise model reduction in circuit QED". arXiv preprint arXiv:2605.03946 (2026).
- [2] **Dirnegger N.**, Narang P., Arora A., "Nonreciprocal quantum information processing with superconducting diodes in circuit quantum electrodynamics", arXiv:2511.20758 (2025), under review Nature npj quantum.
- [3] **Dirnegger, N.**, Arora, A. and Narang, P., Superconducting Diodes for Qubit Readout, U.S. Patent No. 63/904,472. 23 Oct 2025
- [4] **Dirnegger, N.**, Arora, A. and Narang, P., Superconducting Diodes for Qubit-Qubit Coupling, U.S. Patent No. 63/904,486. 23 Oct 2025
- [5] **Dirnegger N.**, Wesson M., Arora A., Petrides I., Curtis J., Been E., Yacoby A., Narang P., "Probing time-reversal symmetry breaking using a nonlinear superconducting ring resonator", Physical Review A, 10.1103
- [6] **Dirnegger. N**, Malekakhlagh M., Siddhu V., Rao A., Xiong C., Kumph M., Orcutt J., Falk A., "Distilled remote entanglement between superconducting qubits across optical channels". Quantum Science and Technology, 11(1), 015026.
- [7] Negrin, R., **Dirnegger, N.**, Munizzi, W., Talukdar, J. and Narang, P., 2025, March. Efficient multiparty entanglement distribution with DODAG-X protocol. In Quantum Computing, Communication, and Simulation V (Vol. 13391, pp. 39-71) (2024). SPIE.

Talks & Presentations

June 2026

New York City, USA

Distilled remote entanglement of superconducting qubits via optical channels, *PA-NYC, Columbia University*

March 2026

Denver, USA

Nonreciprocal Qubit Coupling through superconducting diodes for modular quantum circuits, *APS Physics March Meeting*

January 2026

San Francisco, USA

Extending quantum computing through modularity, *SPIE Photonics West*

November 2025

Berkeley, USA

Nonreciprocal Microwave Quantum Device for Quantum Computing Applications, *Invited Talk at Rigetti Computing*

March 2025

Anaheim, USA

Novel Detection Method of Time Reversal Symmetry Broken Order using a Non-linear Superconducting Ring Resonator, *APS Presentation*

January 2025

San Francisco, USA

Efficient Multiparty Entanglement Distribution with DODAG-X Protocol, *SPIE Photonics West*

August 2024

New York, USA

Remote Entanglement of superconducting qubits across optical channels, *Invited Talk at NYU*

August 2025

Yorktown Heights, USA

Remote Entanglement of superconducting qubits across optical channels, *Invited IBM Talk*

March 2024

Minneapolis, USA

Superconducting Ring Resonators as Sensitive Quantum Sensors, *APS Summit*